

Sports Injury Risk Detection from Video

Simplified Database Schema

This simplified schema is designed for an undergraduate capstone project. It supports all major project features while keeping the database easy to understand and implement.

PostgreSQL Tables

users

Purpose: Stores all platform users and their authentication/profile information.

Attribute	Data Type
user_id	UUID (PK)
name	VARCHAR
email	VARCHAR (Unique)
password	TEXT
role	ENUM
phone	VARCHAR
profile_image	TEXT
created_at	TIMESTAMP

athletes

Purpose: Stores athlete-specific information, physical assessment and training profile.

Attribute	Data Type
athlete_id	UUID (PK)
user_id	UUID (FK)
sport	VARCHAR
position	VARCHAR
age	INT
height	FLOAT
weight	FLOAT
training_load	FLOAT
flexibility	FLOAT
strength	FLOAT
balance	FLOAT
endurance	FLOAT
coach_notes	TEXT

injury_history

Purpose: Maintains historical injury records for each athlete.

Attribute	Data Type
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injury_id	UUID (PK)
athlete_id	UUID (FK)
injury_type	VARCHAR
body_part	VARCHAR
severity	VARCHAR
injury_date	DATE
recovery_date	DATE
remarks	TEXT

videos

Purpose: Stores uploaded videos and processing metadata.

Attribute	Data Type
video_id	UUID (PK)
athlete_id	UUID (FK)
activity	VARCHAR
video_url	TEXT
duration	FLOAT
fps	INT
resolution	VARCHAR
quality_score	FLOAT
processing_status	VARCHAR
uploaded_at	TIMESTAMP

analysis_results

Purpose: Stores biomechanical measurements, movement analysis and overall risk summary.

Attribute	Data Type
analysis_id	UUID (PK)
video_id	UUID (FK)
athlete_id	UUID (FK)
knee_valgus	FLOAT
hip_stability	FLOAT
trunk_lean	FLOAT
stride_length	FLOAT
joint_alignment	FLOAT
symmetry_score	FLOAT
fatigue_score	FLOAT
movement_quality	FLOAT
overall_risk_score	FLOAT
risk_level	VARCHAR
created_at	TIMESTAMP

injury_predictions

Purpose: Stores predicted probabilities for supported injury categories.

Attribute	Data Type
prediction_id	UUID (PK)
analysis_id	UUID (FK)
acl_risk	FLOAT
hamstring_risk	FLOAT
ankle_risk	FLOAT
shoulder_risk	FLOAT
lower_back_risk	FLOAT
overuse_risk	FLOAT

recommendations

Purpose: Stores AI-generated corrective and recovery recommendations.

Attribute	Data Type
recommendation_id	UUID (PK)
prediction_id	UUID (FK)
exercise	TEXT
mobility	TEXT
strengthening	TEXT
recovery	TEXT
training_modification	TEXT

notifications

Purpose: Stores alerts and notifications sent to users.

Attribute	Data Type
notification_id	UUID (PK)
user_id	UUID (FK)
title	VARCHAR
message	TEXT
notification_type	VARCHAR
is_read	BOOLEAN
created_at	TIMESTAMP

reports

Purpose: Stores generated reports and exported documents.

Attribute	Data Type
report_id	UUID (PK)
athlete_id	UUID (FK)
report_type	VARCHAR
generated_by	UUID (FK)

file_path	TEXT
generated_at	TIMESTAMP

performance_records

Purpose: Stores athlete performance history for trend analysis.

Attribute	Data Type
record_id	UUID (PK)
athlete_id	UUID (FK)
activity	VARCHAR
score	FLOAT
remarks	TEXT
recorded_at	TIMESTAMP

MongoDB Collections

pose_data

Purpose: Stores frame-wise pose keypoints and skeleton information generated from videos.

- video_id
- athlete_id
- frames[]
- keypoints[]
- skeleton[]

ai_logs

Purpose: Stores AI model execution logs for debugging and evaluation.

- video_id
- model_name
- inference_time
- confidence
- output

Design Summary

This simplified schema consists of 10 PostgreSQL tables and 2 MongoDB collections. It reduces complexity while covering authentication, athlete management, video processing, AI analysis, injury prediction, recommendations, notifications, reports and performance tracking.